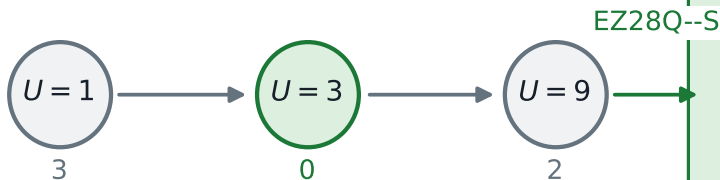


Complete fixed-Q fibre at $p = 41161$, $h = 11$, $j = 3$

$$U \mid j^2 = 9$$



residues of $(p + 4U) \pmod{11}$ beneath the three divisors

Unique fixed-Q target ($U = 3$)

$$R_U = (p + 4U)/h = 3743, \quad A_U = (p + R_U)/4 = 11226$$

$$(h_U, r_U, s_U, \lambda_U) = (3, 1, 3742, 1)$$

$$4/41161 = 1/11226 + 1/462073386 + 1/123483$$

Why this strictly extends the selected square word

$$\alpha = -4c^2 = -100, \text{ so } c = 5 \text{ but } 5 \nmid j = 3.$$

Selected square word unavailable;
nonsquare $U = 3$ still returns.

$$\text{General theorem: } \mathcal{V}_{p,h} = \{U > 0 : U \mid j^2, h \mid p + 4U\}$$

$$\mathcal{V}_{p,h} \leftrightarrow \{ \text{all original M states with } Q = h \}$$

This fibre starts from an existing E state; it does not force occupancy.